
GROUP 02

**Melon AI Learning App
<Iteration/Master> Test Plan**

Version 1.0

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Revision History

Date	Version	Description	Author
22/06/2026	1.0	Initial PA4 test plan filled from Test cases.md, Test report.md, PA4-2026.pdf Section b, and PA3 project documents.	Melon AI team / Tester: Vu

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<Iteration/ Master> Test Plan

1. Introduction

1.1 Purpose

This document defines the test plan for the Melon AI Learning App for Project Assignment 4. It describes the features to be tested, the test environment, tools, team responsibilities, and the current execution summary. It is intended for the project team, instructors, testers, and developers who need to understand what was tested and how test results should be interpreted.

The plan follows the PA4-2026 Section b requirement: prepare a test plan, design test cases, execute implemented test cases, record Pass/Fail status, summarize results, and link failed test cases to reported defects.

1.2 Product Overview

Melon is an AI-assisted learning application for children, parents, and administrators. Children use the system to study lessons, practice exercises, receive AI tutoring support, and track progress. Parents manage child accounts, view progress, set learning preferences, and manage subscription status. Administrators manage users, question banks, and AI parsing workflows.

The current architecture uses a Next.js 16 web application as the frontend and API proxy layer, with a Python FastAPI backend for AI-heavy features such as PDF/OCR parsing, RAG-based question generation, tutoring, and text-to-speech. The system integrates with Firebase Auth/Storage, Stripe test payment flows, vector storage, OpenRouter/OpenAI-compatible LLM services, and ElevenLabs.

1.3 Test Approach and Scope

- Functional, role-based, security, subscription, AI parsing, practice, family-management, administration, and selected UX scenarios are covered.
- Test cases are organized by use-case group and include role, detailed actions, expected result, and execution status.
- Implemented flows were executed in the local development environment; unimplemented flows are marked as skipped/not implemented.
- Failed test cases are linked to defects in the test report so that developers can prioritize remediation before release.
- AI-dependent tests focus on observable contract behavior, returned structured data, and user-facing workflow correctness rather than deterministic model output.

2. Target Test Items

The following use cases/features are the target software items for PA4 testing. Detailed step-by-step test cases are maintained in `pa/pa4/Test cases.md`. The execution status is summarized from `pa/pa4/Test report.md`.

ID	Use Case / Feature Area	Main Roles	Representative Test IDs	Current Status
UC01	Authentication & Security	Kid, Parent, Admin	AUTH-01 to AUTH-09	Executed: 8 pass, 1 fail
UC02	Application Features & Payments	Kid Free, Kid Pro, Parent	SUBS-01 to SUBS-06	Executed: 4 pass, 2 fail
UC03	Knowledge Base & Parsing	Kid Pro, Admin	PARSE-01 to PARSE-05	Executed: 4 pass, 1 fail
UC04	Practice & Personalization	Kid	PRAC-01 to PRAC-05	Executed: 4 pass, 1 fail
UC05	Gamification & System Admin	Kid, Parent, Admin	GAMI-01 to GAMI-05, FAM-01, ADM-01, ADM-04	Executed: 9 pass, 1 fail

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UC06	Bài giảng & Các tính năng khác	Kid, Parent, Admin	to ADM-06 RAG-01 to RAG-02, PROF-01 to PROF-02, ERR-03	Executed: 3 pass, 2 fail
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Note: Out of the 15 Use Cases defined in the Software Architecture Document (PA3), the 6 Core Feature Areas above are implemented and targeted for PA4 testing (MVP). The remaining Use Cases are considered Out of Scope for this iteration and deferred to future releases.

Out of scope for this test plan: production load testing, cross-browser certification, formal accessibility audit, paid live Stripe transactions, and model-quality evaluation. ML model evaluation is handled in the separate PA4 model evaluation deliverable.

3. Environmental Needs

3.1 Hardware Requirements

Hardware Item	Minimum / Recommended Configuration	Purpose
Developer/Test Laptop	Windows machine, 8 GB RAM minimum; 16 GB recommended	Run frontend, backend, browser, and test tools locally
Network Connection	Stable internet connection	Access Firebase, Stripe test mode, OpenRouter/OpenAI-compatible APIs, ElevenLabs, and package registries
Disk Storage	At least 5 GB free space	Store dependencies, uploaded files, generated audio, logs, and local vector data

3.2 Software in the Test Environment

Software Name	Purpose	Version / Configuration	Type and Other Notes
Windows	Local operating system	Windows development environment	Primary local test OS
Node.js and npm	Run Next.js frontend	Compatible with Next.js 16	Frontend dependency/runtime tooling
Next.js Web App	User interface and API proxy	Next.js 16, React, TypeScript, Tailwind CSS	Runs at http://localhost:3000
Python / Conda	Backend runtime	Python 3.11 in conda env myenv	Backend dependency/runtime tooling
FastAPI / Uvicorn	AI backend service	uvicorn on 127.0.0.1:8001	Health endpoint: /health
Firebase Auth and Storage	Identity and object storage	Project test configuration	Used for Kid/Parent/Admin accounts and assets
Stripe Test Mode	Subscription checkout and portal testing	Test cards such as 4242...	No live payment is used
OpenRouter/OpenAI-c	LLM tutoring, quiz	Configured by API keys	External AI

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compatible APIs	generation, OCR/vision workflows		dependency
ElevenLabs	Text-to-speech	Configured by API key	External AI dependency
Chrome/Firefox	Manual UI testing	Current stable browser	Browser DevTools used for inspection

3.3 Productivity and Support Tools

Tool Category or Type	Tool Brand Name	Vendor or In-house	Version / Notes
Source Control	Git	Open source	Track implementation and test artifacts
Editor/IDE	Visual Studio Code	Microsoft	Code and documentation editing
API Testing	Postman / curl / FastAPI Swagger UI	Mixed	Direct API verification and security regression checks
Browser Testing	Chrome DevTools / Firefox DevTools	Google/Mozilla	UI inspection, network tracing, console errors
Automation Support	Project test scripts and manual checklists	In-house	Used to execute and record PA4 test results
Defect Tracking	Test report markdown	In-house	Defects documented in pa/pa4/Test report.md
Payment Dashboard	Stripe Dashboard	Stripe	Verify checkout, portal, webhook behavior in test mode
Cloud Console	Firebase Console	Google	Inspect users, auth state, and storage artifacts

4. Responsibilities, Staffing, and Training Needs

4.1 People and Roles

The test effort assumes that team members can start the local frontend/backend, prepare test users, execute manual/API test cases, and document defects with reproducible steps. External-service keys must be configured before AI, payment, storage, and TTS test cases are executed.

Role	Minimum Resources Recommended	Specific Responsibilities or Comments
Test Manager	1	Plan test scope, coordinate execution, review summary metrics, and ensure PA4 deliverables are complete.
Test Designer	1	Derive test cases from use cases/features, define expected results, and maintain pa/pa4/Test cases.md.
Tester	1-2	Execute manual and API tests, record Pass/Fail/Skipped status, collect evidence, and reproduce defects.
Automation/API Tester	1	Run direct API/security/payment checks with Postman, curl,

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		Swagger UI, or project scripts.
Developer / Defect Owner	2-3	Triage failed test cases, fix code, and provide builds for regression testing.
Test System Administrator	1	Prepare local environment, Firebase/Stripe/API keys, seed test accounts, and maintain backend/frontend availability.